MIL-T-46947A(MU)
30 May 1968

SUPERSEDING
MIL-T-46947(MU)
20 September 1963

MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED, M11
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated, M11, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-413 - Visual Inspection Guide for Rubber O-Rings
MIL-STD-1167 - Ammunition Data Cards

DRAWINGS

Munitions Command
F10521566 - Thruster, Cartridge Actuated, M11, Metal Parts Assembly
C8800969 - Container, Ammunition, Fiber M434 for Thruster, Cartridge Actuated, M11 with Cartridge M141

FSC 1377

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DRAWINGS (Cont'd)

Munitions Command

C8800970	- Box, Packing, Ammunition for Thruster, Cartridge Actuated, M11 with Cartridge M141 in Fiber Container M434
IEL-10521566	- Inspection Equipment List for Thruster, Cartridge Actuated M11, Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture, Surface Roughness, Waviness and Lay

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and in the reference specification.

3.2 Material.- The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.3 Mechanical properties.- Mechanical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.4 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing.

3.5 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.6 Residual magnetism.- The metal parts assembly shall not deflect the indicator of a compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

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3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 5 metal parts assemblies, 10 sets of parts, fifty shear screws, fifty shear pins and fifty curved washers from the same lots tested by the contractor. The samples shall be subjected to tests in 4.3.3. The test procedures shall be in accordance with 4.4. The samples shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as used in regular production and shall be obtained from the same source of supply.

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4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- Unless otherwise specified by the contracting officer, formation of lots and presentation of lots shall be in accordance with the "Submission of Product" provisions of MIL-STD-105.

4.3.1.2 Size of lot.- The shipping lot of metal parts assemblies shall consist of 1200 thrusters, maximum, provided the homogeneity requirements of inspection lots have been retained.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and associated criteria in Table I and II. For an individual defect (single characteristic), the criteria in Table I shall be used. For total defects within a major or minor group, the criteria in Table II shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber o-rings shall be in accordance with MIL-STD-413. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table I

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1,300	100	1	2	2	3
1,301 - 3,200	150	2	3	3	4
3,201 - 8,000	225	3	4	4	5
8,001 - 22,000	300	4	5	5	6

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Table II

Sample Sizes for Major Defects No. of Defects in Class	Total Allowable Defects per Sample Size 110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14

Sample Sizes for Minor Defects No. of Defects in Class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.3.2.1 Body, Dwg. D10521568, covering a detail of Dwg. F10521566.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of groove with large cavity	Gage
102. Concentricity of internal counterbore with large cavity	Gage
103. Concentricity of small cavity with large cavity	Gage
104. Concentricity of body with large cavity	Gage
105. True position of pin hole	Gage
106. Diameter of large cavity	Gage
107. Diameter of groove	Gage
108. Diameter of small cavity	Gage
109. Location of groove, max.	Gage
110. Profile of nose	Gage
111. Metal defective	Visual
112. Protective finish inadequate or defective	Visual

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4.3.2.1 (Cont'd)

	Defect	Method of Inspection
MAJOR		
113.	Surface finish of cavity and nose, improper	Visual
114.	Burr or sharp corners	Visual
115.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread	Gage
202.	Minor diameter of thread, max.	Gage
203.	Diameter of pin hole	Gage
204.	Diameter of body	Gage
205.	Diameter of internal counterbore	Gage
206.	Diameter at intersection of tapers	Gage
207.	Depth of pin hole	Gage
208.	Depth of large cavity	Gage
209.	Depth to face of internal counterbore	Gage
210.	Depth to bottom of counterbore	Gage
211.	Depth to bottom of small cavity	Gage
212.	Depth of effective thread, min.	Gage
213.	Depth of thread, max.	Gage
214.	Width of groove	Gage
215.	Location of groove, min.	Gage
216.	Location of pin hole	Gage
217.	Angle at base of large cavity	Gage
218.	Diameter of counterbore, pin hole	Gage
219.	Depth to drill point	Gage
220.	Length of tip	Gage
221.	Length of nose	Gage
222.	Total length	Gage
223.	Location of taper intersection	Gage
224.	Marking missing, incorrect or illegible	Gage
225.	Radii or chamfer incorrect	Gage
226.	Surface finish improper, except as otherwise classified	Visual

4.3.2.2 Head Dwg. F10521569, covering a detail of Dwg. C10521583.

CRITICAL

None defined

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4.3.2.2 (Cont'd)

	Defect	Method of Inspection
MAJOR		
101.	Pitch diameter of external thread, min.	Gage
102.	Major diameter of external thread, min.	Gage
103.	Pitch diameter of internal thread, max.	Gage
104.	Minor diameter of internal thread, max.	Gage
105.	Concentricity of body with internal thread	Gage
106.	Concentricity of small cavity with internal thread	Gage
107.	Concentricity of counterbore with internal thread	Gage
108.	Concentricity of large cavity with internal thread	Gage
109.	Concentricity of shoulder with internal thread	Gage
110.	Concentricity of internal groove with large cavity	Gage
111.	Concentricity of external groove with shoulder	Gage
112.	True position of pin hole	Gage
113.	Diameter of small cavity	Gage
114.	Diameter of counterbore	Gage
115.	Diameter of large cavity	Gage
116.	Diameter of body	Gage
117.	Diameter of large shoulder, max.	Gage
118.	Diameter of pin hole	Gage
119.	Depth to bottom of internal thread undercut	Gage
120.	Depth to bottom of large cavity, max.	Gage
121.	Depth to bottom of counterbore, max.	Gage
122.	Profile of internal groove	Gage
123.	Profile of external groove	Gage
124.	Metal defective	Visual
125.	Protective finish inadequate or defective	Visual
126.	Surface finish of small cavity and body, improper	Visual
127.	Burr or sharp edge	Visual
128.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of external thread, max.	Gage
202.	Pitch diameter of internal thread, min.	Gage
203.	Diameter of large shoulder, min.	Gage
204.	Diameter of pellet hole in external thread	Gage
205.	Diameter of groove, external	Gage
206.	Diameter of groove, internal	Gage
207.	Diameter of undercut, internal thread	Gage

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MINOR	
208. Diameter of undercut, external thread	Gage
209. Depth to bottom of large cavity, min.	Gage
210. Depth to bottom of small cavity	Gage
211. Depth to bottom of internal groove	Gage
212. Depth to bottom of counterbore, min.	Gage
213. Depth of pellet hole	Gage
214. Length of body	Gage
215. Total length	Gage
216. Width across flats	Gage
217. Location of flats	Gage
218. Location of groove, external	Gage
219. Location of hole thru external thread	Gage
220. Diameter of shoulder at flats	Gage
221. Diameter of hole, external threaded end	Gage
222. Length to end of flats	Gage
223. Length to angle on shoulder	Gage
224. Width of undercut, internal	Gage
225. Width of undercut, external	Gage
226. Location of undercut, external	Gage
227. Marking missing, incorrect or illegible	Visual
228. Radii or chamfer incorrect	Gage
229. Surface finish improper, except as otherwise classified	Visual

4.3.2.3 Retainer, Dwg. D10521570, covering a detail of Dwg. C10521575.

CRITICAL

None defined

MAJOR

101. Pitch diameter of external thread, min.	Gage
102. Major diameter of external thread, min.	Gage
103. Pitch diameter of internal thread, max.	Gage
104. Minor diameter of internal thread, max.	Gage
105. Concentricity of body with external and internal threads	Gage
106. Concentricity of long cavity with short cavity	Gage
107. Concentricity of short cavity with external thread	Gage
108. Concentricity of counterbore with short cavity	Gage

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4.3.2.3 (Cont'd)

Defect		Method of Inspection
MAJOR		
109.	Concentricity of thread lead with external thread	Gage
110.	Diameter of body	Gage
111.	Diameter of thread lead, min.	Gage
112.	Diameter of counterbore, max.	Gage
113.	Diameter of short cavity, min.	Gage
114.	Radii at base of counterbore and mouth of short cavity	Gage
115.	Thread damaged or not full	Visual
116.	Metal defective	Visual
117.	Protective finish inadequate or defective	Visual
118.	Surface finish on body, improper	Visual
119.	Burr or sharp corners	Visual
120.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of external thread, max.	Gage
202.	Pitch diameter of internal thread, min.	Gage
203.	Diameter of thread lead, max.	Gage
204.	Diameter of counterbore, min.	Gage
205.	Diameter of short cavity, max.	Gage
206.	Diameter of long cavity	Gage
207.	Depth of counterbore	Gage
208.	Depth to base of short cavity	Gage
209.	Total length	Gage
210.	Length of body	Gage
211.	Length of thread lead	Gage
212.	Length of effective internal thread, min.	Gage
213.	Marking missing, incorrect or illegible	Visual
214.	Radius or chamfer incorrect except as otherwise classified	Gage
215.	Surface finish improper, except as otherwise classified	Visual

4.3.2.4 Guide, Drawing C10521572, covering a detail of Drawing F10521566

CRITICAL

None defined

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Concentricity of hole with large diameter	Gage
102. Diameter of hole, min.	Gage
103. Large diameter	Gage
104. Thickness, max.	Gage
105. Metal defective	Visual
106. Protective finish inadequate or defective	Visual
107. Surface finish, improper	Visual
108. Burr	Visual
109. Foreign matter	Visual
MINOR	
201. Diameter of hole, max.	Gage
202. Height of boss	Gage
203. Thickness, min.	Gage
204. Diameter of boss	Gage
205. Marking missing, incorrect or illegible	Visual
206. Radius or chamfer incorrect	Gage

4.3.2.5 Pin, Firing, Dwg. C10521573, covering a detail of Dwg. F10521566.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of groove with body	Gage
102. Concentricity of striker with body	Gage
103. True position of hole	Gage
104. Diameter of body, min.	Gage
105. Diameter of groove	Gage
106. Diameter of striker	Gage
107. Diameter of pin hole, max.	Gage
108. Length of striker, min.	Gage
109. Profile of striker	Gage
110. Profile of groove	Gage
111. Metal defective	Visual
112. Shear pin hole edges not sharp	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR	
113. Protective finish inadequate or defective	Visual
114. Surface finish of striker and body, improper	Visual
115. Burr or sharp corners	Visual
116. Foreign matter	Visual
MINOR	
201. Diameter of body, max.	Gage
202. Diameter of pin hole, min.	Gage
203. Depth of slot	Gage
204. Length of striker, max.	Gage
205. Total length	Gage
206. Location of groove	Gage
207. Location of hole	Gage
208. Depth of hole	Gage
209. Width of slot	Gage
210. Width of groove	Gage
211. Location of slot	Gage
212. Marking missing, incorrect or illegible	Visual
213. Radius or chamfer incorrect	Gage
214. Surface finish improper, except as otherwise classified	Visual

4.3.2.6 Pin, Shear, Dwg. C10521574, covering a detail of Dwg. F10521566.

CRITICAL

None defined

MAJOR

101. Diameter, max.	Gage
102. Total length , min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr or sharp corners	Visual
106. Foreign matter	Visual

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4.3.2.6 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Total length, max.	Gage
202.	Surface finish, improper	Visual

4.3.2.7 Ring, Segment, Dwg. C10521576, covering a detail of Dwg. F10521566.

CRITICAL

None defined

MAJOR

101.	Length	Gage
102.	Inside radius	Gage
103.	Outside radius	Gage
104.	Width	Gage
105.	Metal defective	Visual
106.	Protective finish inadequate or defective	Visual
107.	Surface finish, improper	Visual
108.	Burr	Visual
109.	Foreign matter	Visual

MINOR

201.	Marking missing, incorrect or illegible	Visual
202.	Radius incorrect except as otherwise classified	Gage

4.3.2.8 Screw, Shear, Dwg. C10521577, covering a detail of Dwg. F10521566.

CRITICAL

None defined

MAJOR

101.	Diameter of body	Gage
102.	Metal defective	Visual
103.	Protective finish inadequate or defective	Visual
104.	Burr	Visual
105.	Foreign matter	Visual

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4.3.2.8 (Cont'd)

Defect		Method of Inspection
MINOR		
201.	Pitch diameter of thread	Gage
202.	Minor diameter of thread, max.	Gage
203.	Depth of slot	Gage
204.	Length of body	Gage
205.	Width of slot	Gage
206.	Location of slot	Gage
207.	Total length	Gage
208.	Radius or chamfer incorrect	Gage
209.	Surface finish, improper	Visual

4.3.2.9 Tube, Inside, Dwg. C10521578, covering a detail of Dwg. F10521566.

CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of shoulder with body and large cavity	Gage
102.	Concentricity of midsection cavity with large cavity	Gage
103.	Concentricity of bore with large cavity	Gage
104.	Concentricity of groove with bore	Gage
105.	Diameter of groove	Gage
106.	Diameter of bore	Gage
107.	Diameter of midsection cavity	Gage
108.	Diameter of large cavity	Gage
109.	Diameter of body	Gage
110.	Diameter of shoulder	Gage
111.	Profile of groove	Visual
112.	Metal defective	Visual
113.	Protective finish inadequate or defective	Visual
114.	Surface finish of body and shoulder, improper	Visual
115.	Burr or sharp edge	Visual
116.	Foreign matter	Visual
MINOR		
201.	Depth to bore	Gage
202.	Length to shoulder	Gage
203.	Total length	Gage
204.	Location of groove	Gage

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4.3.2.9 (Cont'd)

	Defect	Method of Inspection
MINOR		
205.	Length of midsection cavity	Gage
206.	Width of groove	Gage
207.	Marking missing, incorrect or illegible	Visual
208.	Radius or chamfer incorrect	Gage
209.	Surface finish improper, except as otherwise classified	Visual

4.3.2.10 Tube, Outside, Dwg. C10521579, covering a detail of Dwg. F10521566.

CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of bore with cavity	Gage
102.	Concentricity of internal groove with bore	Gage
103.	Concentricity of shoulder with body and cavity	Gage
104.	Concentricity of external groove with shoulder	Gage
105.	Diameter of bore	Gage
106.	Diameter of internal groove	Gage
107.	Diameter of cavity	Gage
108.	Diameter of shoulder	Gage
109.	Diameter of external groove	Gage
110.	Diameter of body	Gage
111.	Angle of shoulder	Gage
112.	Profile of internal groove	Gage
113.	Profile of external groove	Gage
114.	Metal defective	Visual
115.	Protective finish inadequate or defective	Visual
116.	Surface finish, improper	Visual
117.	Burr or sharp edge	Visual
118.	Foreign matter	Visual
MINOR		
201.	Depth to bottom of bore	Gage
202.	Length of angle	Gage
203.	Total length	Gage
204.	Length to shoulder	Gage

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4.3.2.10 (Cont'd)

Defect	Method of Inspection
MINOR	
205. Width of internal groove	Gage
206. Width of external groove	Gage
207. Location of internal groove	Gage
208. Location of external groove	Gage
209. Marking missing, incorrect or illegible	Visual
210. Radii or chamfer incorrect	Gage

4.3.2.11 Washer, curved, Dwg. C10521580, covering a detail of Dwg. F10521566.

CRITICAL	
None defined	
MAJOR	
101. Load at specified height	Gage
102. Thickness	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter, outside	Gage
202. Diameter of hole	Gage
203. Height	Gage
204. Surface finish, improper	Visual

4.3.2.12 Stop, Dwg. C10521581, covering a detail of Dwg. C10521582.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of groove with body	Gage
104. Concentricity of body with thread	Gage

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4.3.2.12 (Cont'd)

	Defect	Method of Inspection
MAJOR		
105.	Concentricity of shoulder with thread	Gage
106.	Diameter of shoulder	Gage
107.	Diameter of body	Gage
108.	Thread damaged or not full	Visual
109.	Metal defective	Visual
110.	Protective finish inadequate or defective	Visual
111.	Surface finish of body and shoulder, improper	Visual
112.	Burr	Visual
113.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread, max.	Gage
202.	Diameter of undercut	Gage
203.	Diameter of groove	Gage
204.	Total length	Gage
205.	Length of body	Gage
206.	Width of groove	Gage
207.	Width of flats	Gage
208.	Location of flats	Gage
209.	Location of groove	Gage
210.	Location of undercut	Gage
211.	Diameter of cavity	Gage
212.	Depth of cavity	Gage
213.	Width of undercut	Gage
214.	Marking missing, incorrect or illegible	Visual
215.	Rasius or chamfer incorrect	Gage
216.	Surface finish improper, except as otherwise classified	Visual

4.3.2.13 Retainer and Pellet Assembly, Dwg. C10521575, covering a detail of Dwg. F10521566.

CRITICAL		
	None defined	
MAJOR		
101.	Protrusion of pellet over thread	Gage
102.	Pellet loose, missing or improperly inserted	Visual

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4.3.2.13 (Cont'd)

Defect	Method of Inspection
MAJOR	
103. Thread damaged or not full	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Location of pellets	Measure

4.3.2.14 Stop and Pellet Assembly, Dwg. C10521582, covering a detail of Dwg. F10521566.

CRITICAL

None defined

MAJOR	
101. Protrusion of pellet over thread	Gage
102. Pellet loose, missing or improperly inserted	Visual
103. Thread damaged or not full	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Location of pellets	Gage

4.3.2.15 Head and Pellet Assembly, Dwg. C10521583, covering a detail of Dwg. F10521566.

CRITICAL

None defined

MAJOR	
101. Diameter across top of pellets	Gage
102. Pellet loose, missing or improperly inserted	Visual
103. Thread damaged or not full	Visual
104. Burr	Visual
105. Foreign matter	Visual

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4.3.2.15 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Location of pellets	Gage

4.3.2.16 Metal Parts Assembly, Dwg. F10521566, less all components except Guide 10521572 and Firing Pin 10521573.

CRITICAL	
1. Protrusion of firing pin	Gage
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.3.2.17 Metal Parts Assembly, Dwg. F10521566.

CRITICAL	
None defined	
MAJOR	
101. Pellet missing in shear pin hole	Visual
102. Pellet missing in head threads	Visual
103. Improper assembly	Visual
104. Any part missing, not otherwise classified	Visual
105. Material defective	Visual
MINOR	
201. Protective finish defective	Visual
202. Shipping cap missing	Visual
203. Foreign matter	Visual

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4.3.2.18 Fiber Container, Dwg. C8800969.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Metal ends loose or distorted	Manual
203. Gap between body and cover, max.	Visual
204. Cut or damaged through all impregnated layers	Visual

4.3.2.19 Packing Box, Dwg. C8800970.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual & Visual
203. Strapping missing, loose, broken or improperly assembled	Manual & Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

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4.3.3.1 Hardness.- Fifty units from each heat treated lot of components shall be tested for compliance with the values noted on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.2 Shear strength.

4.3.3.2.1 Shear pin.- Fifty pins from each lot shall be tested for shear strength. Failure of any pin to comply with the requirements of the applicable drawing shall be cause for rejection of the lot.

4.3.3.2.2 Shear screw.- Fifty screws from each lot shall be tested for shear strength. Failure of any screw to comply with the requirements of the applicable drawing shall be cause for rejection of the lot.

4.3.3.3 Curved washers.- Fifty curved washers shall be tested for hydrogen embrittlement. Failure of any washer to pass the test shall be cause for rejection of the lot.

4.3.3.4 Protective finish.- The number of units for protective finish tests shall be in accordance with the applicable MIL-Standard. Failure of any sample to comply with the requirements of the applicable MIL Standard shall be cause for rejection of the lot.

4.3.3.5 Residual magnetism.- Each assembly shall be subjected to a test for excessive residual magnetism. Failure of any unit to comply with the requirements of 3.6 shall be cause for rejection of the assembly.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10521566.

4.4 Test methods and procedures.

4.4.1 Hardness.- Prior to the application of protective finish, a hardness tester shall be used to determine the hardness values specified on the detail drawing.

4.4.2 Shear strength.

4.4.2.1 Shear pin.- After application of protective finish, the load shall be applied at the steady rate of 0.125 inches per minute.

4.4.2.2 Shear screw.- After application of protective finish, the load shall be applied at the steady rate of 0.125 inches per minute.

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4.4.3 Curved washers.- After application of protective finish, the washers shall be kept in a fully compressed position for a three hour period followed with a visual inspection for cracks.

4.4.4 Protective finish.- Test procedures for protective finish shall be in accordance with the provisions referenced on the component drawing.

4.4.5 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster 6 inches from, and at the same height as, the compass in a north-south horizontal position with the compass set in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Level A - Packing and marking shall be in accordance with Drawings C8800969 and C8800970.

5.2 Data cards.- Data card information shall be as specified in MIL-A-2550.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A240

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